Coordinate Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Review: Circumference and area**

In the diagram below, two chords, AB and CD, intersect at point P inside a circle. If $AP = 5$ cm, $PB = 8$ cm, and $CP = 4$ cm, what is the length of PD in cm?

According to the Power of a Point theorem for intersecting chords, the product of the segments of one chord is equal to the product of the segments of the other chord. In this case, $AP \cdot PB = CP \cdot PD$. We are given $AP = 5$ cm, $PB = 8$ cm, and $CP = 4$ cm. Substituting these values into the equation, we get $5 \cdot 8 = 4 \cdot PD$. This simplifies to $40 = 4 \cdot PD$. Dividing both sides by 4, we find $PD = 10$ cm. Therefore, the length of PD is 10 cm.

2 **[Foundational]** • **Target: Arc length and sector area**

In the figure below with $AB \parallel CD$, $\text{Area}(\Delta ABE) = 36$ and $\text{Area}(\Delta CDE) = 81$. If base $AB = 10$, what is the length of base CD?

Because $\Delta ABE \sim \Delta CDE$, the ratio of their areas equals the square of the linear scale factor: $\frac{\text{Area}(\Delta CDE)}{\text{Area}(\Delta ABE)} = \frac{81}{36} = \frac{9}{4} = k^2$. Taking the square root gives $k = \sqrt{\frac{9}{4}} = \frac{3}{2}$. Therefore, $CD = k \cdot AB = \frac{3}{2} \times 10 = 15$.

3 **[Foundational]** • **Target: Extension: Chords of a circle**

The triangle ABC is translated on the coordinate plane according to the rule $(x, y) \to (x - 3, y + 2)$. If the vertices of the image triangle A'B'C' are A'(-1, 5), B'(3, 1), and C'(-2, -3), what are the coordinates of the original vertices A, B, and C?

The translation rule $(x, y) \to (x - 3, y + 2)$ means that to find the image coordinates (x', y') from the pre-image coordinates (x, y), we subtract 3 from the x-coordinate and add 2 to the y-coordinate: $x' = x - 3$ and $y' = y + 2$. To find the pre-image coordinates from the image coordinates, we reverse this process. This means we add 3 to the x-coordinate of the image and subtract 2 from the y-coordinate of the image: $x = x' + 3$ and $y = y' - 2$. For vertex A': $x = -1 + 3 = 2$, $y = 5 - 2 = 3$..

4 **[Foundational]** • **Target: Inscribed angles and polygons**

In ΔABC, $AD = 3$ and $DB = 9$ (so $AB = 12$); $AE = 4$ and $EC = 5$ (so $AC = 9$). If base $BC = 21$, what is the length of segment DE?

Compare the side ratios across the shared angle $\angle A$: $\frac{AD}{AC} = \frac{3}{9} = \frac{1}{3}$ and $\frac{AE}{AB} = \frac{4}{12} = \frac{1}{3}$. Because two pairs of corresponding sides are in the same constant ratio $1/3$ and include the shared angle $\angle A$, $\Delta ADE \sim \Delta ACB$ by SAS similarity in inverted order. Thus, $\frac{DE}{BC} = \frac{1}{3} \Rightarrow DE = 21 \times \frac{1}{3} = 7$.

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5 [Foundational] • Target: Angles from intersecting chords, secants, and tangents

The diagonals of trapezoid $ABCD$ with $AB \parallel CD$ intersect at P. If $\text{Area}(\Delta APB) = 16$ and $\text{Area}(\Delta CPD) = 49$, what is the total area of trapezoid $ABCD$?

Because $\Delta APB \sim \Delta CPD$, the ratio of areas is $16/49 = (4/7)^2$. For any trapezoid, the side triangles have equal area given by the geometric mean of the base triangles: $\text{Area}(\Delta APD) = \text{Area}(\Delta BPC) = \sqrt{16 \times 49} = 4 \times 7 = 28$. The total area of the trapezoid is $16 + 49 + 28 + 28 = 121$ (which equals $(4 + 7)^2 = 11^2 = 121$).

6 [Foundational] • Target: Tangents to a circle

In ΔABC, segment $DE \parallel BC$ with D on AB and E on AC. If $AD = 5$, $DB = 10$, and $DE = 7$, what is the length of base BC?

Because $DE \parallel BC$, $\Delta ADE \sim \Delta ABC$ by AA. The parallel bases correspond to full triangle sides from vertex A: $AB = AD + DB = 5 + 10 = 15$. Setting up the whole-to-part ratio: $\frac{DE}{BC} = \frac{AD}{AB} \Rightarrow \frac{7}{BC} = \frac{5}{15} = \frac{1}{3} \Rightarrow BC = 7 \times 3 = 21$. Note: Using part-to-part $5/10$ gives the distractor 14.

7 [Foundational] • Target: Review: Circumference and area

In the hourglass figure below, segment $AB \parallel CD$, and segments AC and BD intersect at E. If $AB = 8$, $CD = 12$, and $AE = 6$, what is the length of segment CE?

Because $AB \parallel CD$, alternate interior angles $\angle A \cong \angle C$ and $\angle B \cong \angle D$ combined with vertical angles at E prove that $\Delta ABE \sim \Delta CDE$. Setting up the ratio of corresponding sides: $\frac{AE}{CE} = \frac{AB}{CD} \Rightarrow \frac{6}{CE} = \frac{8}{12} = \frac{2}{3} \Rightarrow CE = \frac{6 \times 3}{2} = 9$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arc length and sector area

A right triangle has legs of lengths 15 and 20. What is the exact length of the altitude drawn perpendicular to the hypotenuse?

First compute the hypotenuse using the Pythagorean theorem: $c = \sqrt{15^2 + 20^2} = \sqrt{225 + 400} = \sqrt{625} = 25$. Equating the area calculated in two ways: $\text{Area} = \frac{1}{2}ab = \frac{1}{2}ch \Rightarrow ab = ch$. Substituting: $15 \times 20 = 25 \cdot h \Rightarrow 300 = 25h \Rightarrow h = \frac{300}{25} = 12$.

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9 [Intermediate] • Target: Extension: Chords of a circle

In ΔABC, $\angle ADE \cong \angle C$ with $AD = 5$ and $AC = 15$. If the total area of ΔABC is 108, what is the area of quadrilateral $BDEC$?

Because $\Delta ADE \sim \Delta ACB$, the linear ratio is $k = \frac{AD}{AC} = \frac{5}{15} = \frac{1}{3}$. The area ratio is $k^2 = (1/3)^2 = 1/9$. Thus, $\text{Area}(\Delta ADE) = \frac{1}{9} \times 108 = 12$. The area of quadrilateral $BDEC = \text{Area}(\Delta ABC) - \text{Area}(\Delta ADE) = 108 - 12 = 96$.

10 [Intermediate] • Target: Inscribed angles and polygons

In ΔABC shown below, segment $DE \parallel BC$. If $AD = 6$, $DB = 9$, and $AE = 8$, what is the length of segment EC?

By the Triangle Proportionality Theorem (Thales' Theorem), a line parallel to one side of a triangle divides the other two sides proportionally: $\frac{AD}{DB} = \frac{AE}{EC}$. Substituting the given values: $\frac{6}{9} = \frac{8}{EC} \Rightarrow \frac{2}{3} = \frac{8}{EC} \Rightarrow EC = \frac{3 \times 8}{2} = 12$.

11 [Intermediate] • Target: Angles from intersecting chords, secants, and tangents

In ΔABC, $DE \parallel BC$ with D on AB and E on AC. If $AD = 4$ and $AB = 10$, what fraction of the area of ΔABC is occupied by trapezoid $BDEC$?

Since $DE \parallel BC$, $\Delta ADE \sim \Delta ABC$ with linear scale factor $k = \frac{AD}{AB} = \frac{4}{10} = \frac{2}{5}$. The ratio of the area of ΔADE to the area of ΔABC is $k^2 = (2/5)^2 = 4/25$. The area of trapezoid $BDEC$ is the remaining fraction: $1 - \frac{4}{25} = \frac{21}{25}$.

12 [Intermediate] • Target: Tangents to a circle

Trapezoid $ABCD$ has parallel bases $AB = 6$ and $CD = 14$. The diagonals AC and BD intersect at point P. What is the ratio of $\text{Area}(\Delta APB)$ to $\text{Area}(\Delta CPD)$?

Because $AB \parallel CD$, alternate interior angles and vertical angles prove that $\Delta APB \sim \Delta CPD$. The linear scale factor is $k = \frac{AB}{CD} = \frac{6}{14} = \frac{3}{7}$. The ratio of their areas equals the square of the linear scale factor: $k^2 = \left(\frac{3}{7}\right)^2 = \frac{9}{49}$.

13 [Intermediate] • Target: Review: Circumference and area

In ΔABC, point D lies on AB and E on AC such that $\angle AED \cong \angle B$. If $AE = 6$, $AD = 8$, $AC = 18$, and base $BC = 27$, what is the length of segment DE?

Since $\angle A$ is shared and $\angle AED \cong \angle B$, $\Delta AED \sim \Delta ABC$ by AA similarity. The ratio of corresponding sides is $\frac{DE}{BC} = \frac{AD}{AC}$. Substituting the known lengths: $\frac{DE}{27} = \frac{8}{18} = \frac{4}{9} \Rightarrow DE = 27 \times \frac{4}{9} = 12$.

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14 [Intermediate] • Target: Arc length and sector area

In trapezoid $ABCD$ with $AB \parallel CD$, $AB = 8$ and $CD = 12$. The diagonals intersect at P. If the entire diagonal $AC = 30$, what is the length of segment AP?

Because $\Delta APB \sim \Delta CPD$, corresponding segments along the diagonal satisfy $\frac{AP}{PC} = \frac{AB}{CD} = \frac{8}{12} = \frac{2}{3}$. The total length is $AC = AP + PC = 30$. Thus, $AP = 30 \times \frac{2}{2+3} = 30 \times \frac{2}{5} = 12$ (and $PC = 18$).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Extension: Chords of a circle • Cross-Lesson: 5.01 Right Triangles & Trigonometry

In the bowtie figure with $AB \parallel CD$, segments AC and BD intersect at E. If $AB = 10$, $CD = 15$, and the entire segment $BD = 35$, what is the length of segment BE?

Since $AB \parallel CD$, $\Delta ABE \sim \Delta CDE$. The ratio of corresponding sides is $\frac{BE}{ED} = \frac{AB}{CD} = \frac{10}{15} = \frac{2}{3}$. The total length $BD = BE + ED = 35$. Therefore, $BE = 35 \times \frac{2}{2+3} = 35 \times \frac{2}{5} = 14$.

16 [Challenging] • Target: Inscribed angles and polygons • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC, ray AD bisects $\angle A$ with point D on side BC. If $AB = 8$, $AC = 12$, and $BD = 6$, what is the length of segment DC?

By the Angle Bisector Theorem, an angle bisector divides the opposite side into segments proportional to the adjacent sides: $\frac{BD}{DC} = \frac{AB}{AC}$. Substituting: $\frac{6}{DC} = \frac{8}{12} = \frac{2}{3} \Rightarrow 2 \cdot DC = 18 \Rightarrow DC = 9$.

17 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 7.01 Circles

In ΔABC, ray AD bisects $\angle A$. If $AB = x$, $AC = x + 4$, $BD = 4$, and $DC = 6$, what is the length of side AC?

Apply the Angle Bisector Theorem: $\frac{BD}{DC} = \frac{AB}{AC} \Rightarrow \frac{4}{6} = \frac{x}{x+4} \Rightarrow \frac{2}{3} = \frac{x}{x+4}$. Cross-multiplying: $2(x + 4) = 3x \Rightarrow 2x + 8 = 3x \Rightarrow x = 8$. Side $AC = x + 4 = 8 + 4 = 12$.

18 [Challenging] • Target: Tangents to a circle • Cross-Lesson: 8.01 Area, Surface Area & Volume

In right ΔABC, altitude CD to hypotenuse AB measures 6, and the entire hypotenuse $AB = 13$. What is the length of the longer of the two segments into which D divides AB?

Let the two segments be p and q. We know $p + q = 13$ and $p \cdot q = CD^2 = 6^2 = 36$. These are the roots of the quadratic equation $t^2 - 13t + 36 = 0$. Factoring: $(t - 4)(t - 9) = 0$. The segments are 4 and 9. The longer segment is 9.

19 [Challenging] • Target: Review: Circumference and area • Cross-Lesson: 9.01 Transformations & Symmetry

In ΔABC, segment $DE \parallel BC$. If $AD = 8$, $DB = 4$, and $AC = 18$, what is the length of segment AE?

Total length $AB = AD + DB = 8 + 4 = 12$. The ratio of AD to AB is $\frac{AD}{AB} = \frac{8}{12} = \frac{2}{3}$. By similarity, $\frac{AE}{AC} = \frac{AD}{AB} = \frac{2}{3}$. Therefore, $AE = 18 \times \frac{2}{3} = 12$.

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20 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 1.01 Foundations & Reasoning

In ΔABC, ray AD bisects $\angle A$ with D on BC. If $AB = 10$, $AC = 15$, and the entire base $BC = 20$, what is the length of segment BD?

By the Angle Bisector Theorem: $\frac{BD}{DC} = \frac{AB}{AC} = \frac{10}{15} = \frac{2}{3}$. The total base is $BC = BD + DC = 20$. Therefore, $BD = 20 \times \frac{2}{2+3} = 20 \times \frac{2}{5} = 8$ (and $DC = 12$).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Review: Circumference and area

For the model $f(x) = 6x + 6$ in review: circumference and area, evaluate $f(2)$.

a. 18 ✓

b. 24

c. 12

d. 22

Substitute $x = 2$: $f(2) = 6(2) + 6 = 18$.

22 [Foundational] • Target: Review: Circumference and area

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

23 [Foundational] • Target: Review: Circumference and area

For the model $f(x) = 6x + 6$ in review: circumference and area, evaluate $f(5)$.

a. 36 ✓

b. 42

c. 30

d. 40

Substitute $x = 5$: $f(5) = 6(5) + 6 = 36$.

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24 [Foundational] • Target: Arc length and sector area

For the model $f(x) = 2x + 7$ in arc length and sector area, evaluate $f(2)$.

a. 11 ✓

b. 13

c. 4

d. 15

Substitute $x = 2$: $f(2) = 2(2) + 7 = 11$.

25 [Foundational] • Target: Arc length and sector area

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

26 [Foundational] • Target: Arc length and sector area

For the model $f(x) = 2x + 7$ in arc length and sector area, evaluate $f(5)$.

a. 17 ✓

b. 19

c. 10

d. 21

Substitute $x = 5$: $f(5) = 2(5) + 7 = 17$.

27 [Foundational] • Target: Extension: Chords of a circle

For the model $f(x) = 3x + 8$ in extension: chords of a circle, evaluate $f(2)$.

a. 14 ✓

b. 17

c. 6

d. 18

Substitute $x = 2$: $f(2) = 3(2) + 8 = 14$.

28 [Foundational] • Target: Extension: Chords of a circle

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Review: Circumference and area**

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ **and** $x = 6$ ✓

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

30 **[Intermediate]** • **Target: Review: Circumference and area**

For the model $f(x) = 6x + 6$ in review: circumference and area, evaluate $f(3)$.

a. 24 ✓

b. 30

c. 18

d. 28

Substitute $x = 3$: $f(3) = 6(3) + 6 = 24$.

31 **[Intermediate]** • **Target: Review: Circumference and area**

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

32 **[Intermediate]** • **Target: Arc length and sector area**

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

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33 [Intermediate] • Target: Arc length and sector area

For the model $f(x) = 2x + 7$ in arc length and sector area, evaluate $f(3)$.

a. 13 ✓

b. 15

c. 6

d. 17

Substitute $x = 3$: $f(3) = 2(3) + 7 = 13$.

34 [Intermediate] • Target: Arc length and sector area

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

35 [Intermediate] • Target: Extension: Chords of a circle

A data set recorded for extension: chords of a circle consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

36 [Intermediate] • Target: Extension: Chords of a circle

For the model $f(x) = 3x + 8$ in extension: chords of a circle, evaluate $f(4)$.

a. 20 ✓

b. 23

c. 12

d. 24

Substitute $x = 4$: $f(4) = 3(4) + 8 = 20$.

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37 [Intermediate] • Target: Extension: Chords of a circle

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Inscribed angles and polygons • Cross-Lesson: 8.01 Area, Surface Area & Volume

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 15

c. 13 ✓

d. 17

Substituting $x = 1$ into $y = 4x + 9$ gives $y = 4(1) + 9 = 13$.

39 [Challenging] • Target: Inscribed angles and polygons • Cross-Lesson: 9.01 Transformations & Symmetry

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 13

b. 19

c. 17 ✓

d. 21

Substituting $x = 2$ into $y = 4x + 9$ gives $y = 4(2) + 9 = 17$.

40 [Challenging] • Target: Inscribed angles and polygons • Cross-Lesson: 1.01 Foundations & Reasoning

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm ✓

c. 30 cm

d. 96 cm

Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

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41 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 11.01 Probability

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

42 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 2.01 Lines & Angles

In ΔABC studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85° ✓

b. 95°

c. 77°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 52^\circ) = 85^\circ$.

43 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 3.01 Triangles & Congruence

In ΔABC studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 78° ✓

b. 88°

c. 70°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 59^\circ) = 78^\circ$.

44 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm ✓

c. 42 cm

d. 175 cm

Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

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45 [Challenging] • Target: Tangents to a circle • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A data set recorded for tangents to a circle consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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